Inverse-Reinforcement-Learning-Based Robotic Ultrasound Active Compliance Control in Uncertain Environments

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***Abstract*—Robotic ultrasound systems (RUSs) have gained increasing attention because they can automate repetitive procedures and relieve operators' workloads. However, the complexity and uncertainty of the human surface pose a challenge for stable scanning control. This article proposes a general active compliance control strategy based on inverse reinforcement learning (IRL) to perform adaptable scanning for uncertain and unstructured environments. We analyze the manual scanning process pattern and propose a velocity-and-force-related control strategy to achieve variable force control and handle unpredictable deformation. Then, a hybrid policy optimization framework is proposed to improve transferability. In this framework, a reinforcement learning policy with a predefined reward is built to establish the relationship between contact force and posture. Furthermore, the policy is re-optimized using IRL and generated demonstrations for IRL training. The policy is trained on simple standard phantoms and further evaluated for stability and transferability in unseen and complex environments. Quantitative results show that the difference between the proposed method and the three-dimensional (3-D) reconstructed model in terms of posture is (2.3±1.3°, 1.9±1.2°) in continuous scans. Overall, our method provides a solution for improving the usability of RUSs in real-world environments.**

***Index Terms*—Active compliance control, inverse reinforcement learning (IRL), robotic ultrasound system (RUS), variable force control.**

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I. Introduction

AUTOMATED robotic ultrasound systems (RUS) can effectively improve the efficiency of ultrasound imaging processes and have received much attention in recent years [1]. The purpose of an RUS is to automatically perform scanning to collect ultrasound images of the target, which means that the probe needs to be in brief or constant contact with a soft surface. The complexity and uncertainty of the unstructured environment pose significant challenges for accurate and stable robot-human interaction, which has led to the fact that many currently reported RUS lack the transferability of flexible control in complex environments with varying dynamics [2], [3].

The visual-based method is the primary control mode for the current RUS [4], [5]. Some early researches proposed integrating visual markers to generate the spatial position of the target and further contact the scanned surface [6]. This system required a specifically designed optical marker, a corresponding tracking device, and a structured application environment. Further, to enhance the scenario information and improve control accuracy, some researchers have used RGB-D or stereo cameras to collect depth information for generating the posture based on recognition or segmentation methods [7], [8]. However, the vision-based framework has common issues, such as visual occlusion and complex calibration processes. Especially in RUS scanning tasks, the occlusion of the contact area is unavoidable [9]. In addition, the unpredictable characteristics and deformation also make the visual information insufficient for stable active control. To acquire sufficient information about the tissue, ultrasound images are also used as the primary factor to guide the probe [10], relying on the correct interpretation of the detailed ultrasound images. Overall, the position-based methods are effective in generating the preoperative trajectories but challenging in handling the deformation caused by contacts.

The integration of contact force into posture generation has also been studied to improve the coupling between the probe and the skin. Some classical force control strategies, such as impedance control and proportional integral derivative control, have been used to perform flexible contact [11]. The force information is fused with the scene images or ultrasound images to apply a fixed force vector on the surface [2], [3], [12]. The above approaches still rely on accurate modeling of the environment. Conventional force control approaches also require

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manual adjustment of the force value or controller parameters for specific environments, while the posture adjustments are still derived from the vision information [13]. A combination of conventional force control methods can improve the feasibility of continuous scanning of the RUS, but the complexity of the natural environment limits its application.

For contact-rich tasks in dynamic environments, learning-based force control methods have been extensively studied with the development of learning-based techniques [14]. Khansari et al. [15] proposed a control strategy inspired by human strategies to enable the robot to make surface-surface contact between a handheld object and the environment and validate it on different surfaces. By combining classical force control methods and learning-based approaches, Rozo et al. [16] and Li et al. [17] propose to learn from human demonstrations for impedance control and show high adaptive and learning ability in different tasks and complex environments. In addition to imitation learning, variable impedance control methods using manually defined reward functions and reinforcement learning (RL) models have been proposed to automatically adjust output forces or controller parameters [18]. All the researches provide a reference to improve the stability of force control in uncertain environments. Inspired by the above results, our previous study explored the image-based RL method to follow a soft-moving target [19]. Furthermore, we proposed a force-based RL method to scan a complex phantom [20]. However, these methods are entirely based on preformulated reward functions and still have limitations in terms of transferability in the face of different environments.

Inverse reinforcement learning (IRL) seeks to learn a task's reward or cost function from expert demonstration data and further re-optimize a new policy under the learned-reward role [21]. Compared to other imitation learning methods, such as behavioral cloning [22], [23], IRL has better robustness and transferability in different environments due to variable learned rewards [24]. The advantage of IRL is the potential to use data recorded during task execution and build agents that can imitate experts without interfering with the task process. Compared to RL, which requires manual formulation of reward functions, IRL has better applicability in situations where the environment is difficult to define or model accurately, such as autonomous driving, robot navigation, and contact-rich manipulation [25]. Finn et al. [26] and Fu et al. [27] propose a sampling-based IRL approach and an adversarial generative network-based adversarial IRL (AIRL) approach to improve the performance of a single state-space pair. Based on this approach, Zhang et al. [28] verify the feasibility of learning the gain of variable impedance control from expert data and offer a new solution for force control tasks.

The environment uncertainty in the automated RUS scanning task is mainly the deformation and varying stiffness of the skin in contact. The fundamental solution to this challenge is establishing a robust relationship between RUS and the skin. While some related work has attempted visual and learning approaches, it still relies on predefinition and modeling of the environment. The IRL method offers a potential solution by introducing expert knowledge into the model for RUS control in unstructured environments. By analyzing the control pattern during manual scanning in the clinical environment, we propose a force-velocity-based action space and an IRL-based method to perform simultaneous force and posture control for RUS scanning. Our approach uses two parameterized neural networks to establish the relationship between RUS and the contacted skin. First, a simple predefined reward function and RL method establish the relationship between contact force and posture. Meanwhile, a large number of posture trajectories are recorded as expert demonstrations. In the second-round optimization, the policy is re-optimized by a learned reward derived from the generated datasets. Finally, combined with the outer-loop position, the probe continuously scans the human surface. The proposed control strategy for autonomous RUS is shown in Fig. 1. Our main contributions are as follows.

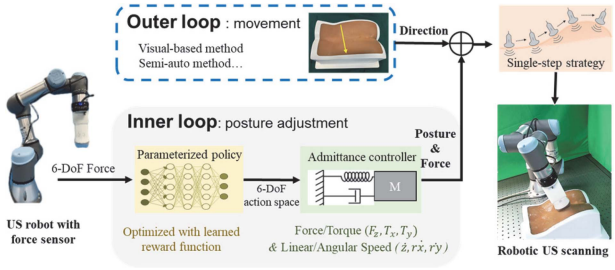


Fig. 1. Robotic ultrasound scanning in our framework is divided into the outer-loop control (path) and the inner-loop control (posture). The proposed method seeks to achieve simultaneous posture and force command output during scanning guided by contact force.

1) We propose a learned-active compliance control framework to perform simultaneous posture and force control for autonomic RUS scanning in an unstructured environment. The motivation of the method is to control the ultrasound probe to maintain a stable coupling to the soft surface guided by contact force information only.
2) The framework contains a force-velocity-based control strategy and a two-round control policy optimization procedure. First, a force-velocity-based action space is proposed to handle uncertain deformations. We introduce expert knowledge by the IRL method to improve stability.
3) The proposed framework can be combined with other outer-loop trajectory/path generation methods for scanning tasks. Experimental results indicate that our framework and policy have satisfactory stability and transferability in different scanning surfaces.

II. Method

A. Force-Velocity-Related Action Space and Control Strategy

The ultrasound scanning task is to collect images by adjusting the relative posture between the probe and the surface. According to the framework shown in Fig. 1, the control loop is divided into outer-loop position control and inner-loop posture control. Here, we focus on the posture control part. For proof of concept, the outer-loop movement position is manually defined in the current work and can be replaced with semi- or automatic procedures.

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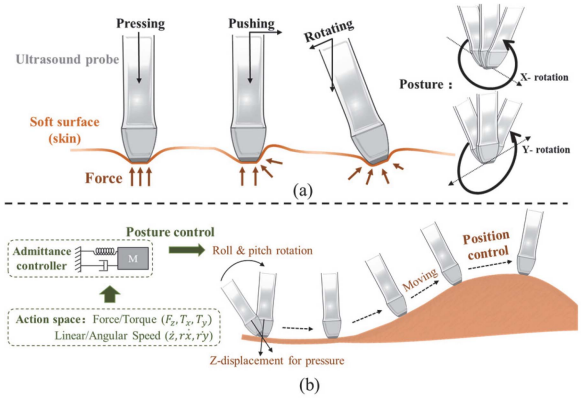


Fig. 2. (a) Surface deformation leads to difficulties in force analysis during ultrasound scanning processes. (b) Discrete single-step control instead of continuous-scan trajectory generation.

As shown in Fig. 2(a), the small contact area between the probe tip and the surface leads to complex surface deformation, making accurate force analysis very difficult. To simplify the analysis and avoid explicit modeling of the entire environment, we propose a discrete single-step posture adjustment strategy instead of complete trajectory or path generation, i.e., the probe reacts based on current force information rather than predicting future actions.

In our strategy, the inner-loop controller performs simultaneous posture and force adjustments. As deformation affects the changes of force and velocity, we propose a 6-degree-of-freedom (DoF) velocity-force-related action space, including the desired contact force $F_{\text{des}} \in \mathbb{R}^1$ in the contact direction (z-direction), the desired torques $T_{\text{des}} \in \mathbb{R}^2$ in roll (Rx) and pitch (Ry) directions, and corresponding linear speed $\dot{x}_{\text{action}}$ and angular speeds ($\dot{r}_{x-\text{action}}, \dot{r}_{y-\text{action}}$). The action is further transformed by an admittance control displacement correction $x_e = x_d - x_0$ in the tool coordinate. According to a standard impedance controller, the dynamic relationship between the correction and the sensed force/torque can be defined by three parameter matrices:

$$F_{\text{ext}} = \boldsymbol{M}\left(\ddot{x}_d - \ddot{x}_0\right) + \boldsymbol{B}\left(\dot{x}_d - \dot{x}_0\right) + \boldsymbol{K}\left(x_d - x_0\right) \tag{1}$$

where $\boldsymbol{M}$ is the inertial matrix, $\boldsymbol{B}$ is the damping matrix, $\boldsymbol{K}$ is the stiffness matrix, x_0 is the initial desired position, x_d is the dynamic desired position and angle, and $F_{\text{ext}} \in \mathbb{R}^3$ is the sensed contact force in the sensor coordinate. Finally, $\dot{x}$ and $\ddot{x}$ are first and second-order derivatives of x, i.e., the velocity and acceleration. Furthermore, according to the principle of the admittance control method, the movement of the robot is adjusted depending on the contact force, and the admittance model in (1) is rewritten as

$$\ddot{x}_e = \boldsymbol{M}^{-1}\left(F_{\text{ext}} - \boldsymbol{B}\dot{x}_e^t - \boldsymbol{K}x_e^t\right) \tag{2}$$

where t is the time step. Integrating the acceleration gives the velocity and position deviation at time step $t+1$

$$\dot{x}_e^{t+1} = \dot{x}_e^t + \ddot{x}_e \Delta t \tag{3}$$

$$x_e^{t+1} = x_e^t + \dot{x}_e^{t+1} \Delta t. \tag{4}$$

Since the admittance controller is equivalent to the movement of the robot as a 6-DoF spring-damped system, the posture adjustment is also applicable to the torque-based admittance control model

$$T_{\text{ext}} = \boldsymbol{M}\ddot{r}_e + \boldsymbol{B}\dot{r}_e + \boldsymbol{K}r_e \tag{5}$$

where r_e is the angular correction in the tool coordinate. Unlike displacement correction, the angular correction r_e needs to be transformed into a posture matrix to be calculated

$$r_e = r_0\, r_d^T \tag{6}$$

$$r_e^{t+1} = \Delta r r^t. \tag{7}$$

From the above equations, the correction x_e^{t+1} is calculated from x_e^t and $\dot{x}_e^t$. When the contact force is constant, x_e^t, $\dot{x}_e^t$ and x_e^{t+1} are generated by the admittance model based on the contact force F_{ext}. However, the changing F_{ext} in the continuous imaging processes means that the generation of x_e^{t+1} is not only based on F_{ext}, but also includes the deviations from $\dot{x}_e^t$ and x_e^t. The removal of deviations is

$$\begin{aligned} F_{\text{ext}} = {} & \boldsymbol{M}\left(\ddot{x}_d - \ddot{x}_0\right) + \boldsymbol{B}\left(\dot{x}_d - \dot{x}_0 - (-\ddot{x}_d\Delta t)\right) \\ & + \boldsymbol{K}\left(x_d - x_0 - (-\dot{x}_d\Delta t)\right). \end{aligned} \tag{8}$$

In general, using the contact force as the action space is sufficient for simple force control tasks as long as the proper controller parameter $[\boldsymbol{M}, \boldsymbol{B}, \boldsymbol{K}]$ is chosen [28]. In ultrasound imaging tasks, the stiffness of the surface in contact with the probe is variable, which means that the generation of action x_e^{t+1} and $\dot{x}_e^{t+1}$ cannot meet the requirement of the changing environment with the static-environment-related parameter matrix. To improve the robustness of RUS to environments with complex stiffness, we choose the action space as the desired contact force F_{action} and the velocity $\dot{x}_{\text{action}}$ in the corresponding direction while avoiding the manually adjusted parameter matrix, which means that the probe's position and velocity are determined by the desired contact force. Meanwhile, the learned velocity of the probe is also force-related

$$\begin{aligned} \ddot{x}_e = {} & \boldsymbol{M}^{-1}\left[\left(F_{\text{action}} - F_{\text{ext}}\right) - \boldsymbol{B}\left(\dot{x}_e^t - (-\ddot{x}_d\Delta t)\right)\right. \\ & \left. - \boldsymbol{K}\left(x_e^t - (-\dot{x}_d\Delta t)\right)\right] \end{aligned} \tag{9}$$

$$\dot{x}_e^{t+1} = \dot{x}_e^t + \dot{x}_{\text{action}} + \ddot{x}_e\Delta t. \tag{10}$$

The correction x_e^{t+1} can be calculated by (4). In ultrasound imaging, this also means that the current state of the robot is independent of the previous action. From robot command execution, the adjustment process of the probe in one position is still relevant at time step $t-1$. Since the action space is velocity-force-related, the three matrices are set here as standard values. As shown in Fig. 2(b), according to the above control strategy, the 6-DoF action is transformed into z displacement, roll, and pitch-rotation, and executed by the robot position controller, which also gives the robot high flexibility in movement. Regarding posture, the ultrasound probe is rotated in robot coordination with the desired 2-DoF torque and the corresponding 2-DoF angular speed. Finally, posture and position are combined to achieve complete probe posture control. Since the posture and position are decoupled, the probe does not need to be calibrated.

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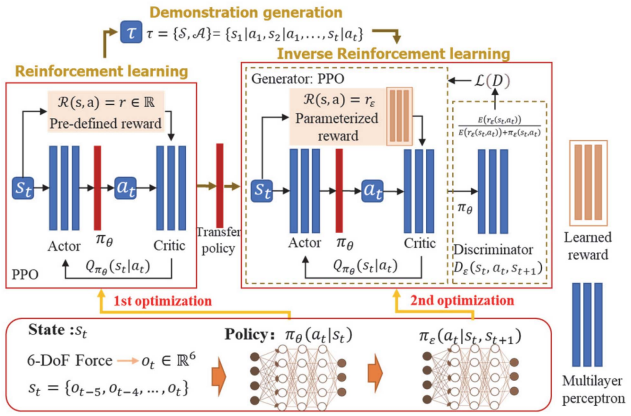


Fig. 3. Policy for posture control is optimized in two rounds. First, the policy is optimized by the predefined reward r. Further, a learned-reward r_ε is used to enhance the transferability and robustness.

B. Hybrid Policy Optimization for Stable Decision Making

Our previous work used the RL method for a similar task [20]. The principle of our method is to use the RL agent and a pre-formulated reward function to minimize the sensed torque. Since the preformulated reward still relies on the manual description and definition of the environment, it has obvious limitations when the test environment differs from the training environment. Inspired [24] and [28], we propose an RL-IRL-based hybrid framework for policy training and stable decision-making, as shown in Fig. 3. The state of the environment $s_t \in \mathbb{R}^{30}$ at time step t is defined as the five continuous 6-DoF force observation $o_t \in \mathbb{R}^6$, which is collected by the sensor for context enhancement. The commands for interacting with the environment are the 6-DoF desired force and velocity presented in Section II-A. Since IRL optimization requires many demonstrations, we first train an RL agent for both demonstrations and policy initialization to avoid the time consumption of demonstration collection. In the Markov decision process framework theory, the RL process can be defined by the sets of states $\mathcal{S} = \{s_1, s_2, \ldots, s_t\}$, the action $\mathcal{A} = \{a_1, a_2, \ldots, a_t\}$, the stochastic dynamics between states for a given action $p(s_{t+1}|s_t,\ a_t)$, the reward function $\mathcal{R}\ (s,\ a) = \ r \in \mathbb{R}$, and the discount factor $\gamma \in [01)$ [29]. The agent interacting with the environment is a policy $\pi_\theta(a|s)$ parameterized by θ and is trained by maximizing the expected reward $\mathcal{R}$. The predefined reward $r \in \mathbb{R}^1$ is the negative value of the Euler angle α_t between the probe and the contacted surface, which means that the probe is encouraged to maintain a perpendicular relationship with the surface and collect ultrasound images. Also, angle and force exceeding the limits will result in critical negative rewards that seek to keep the system safe. To optimize the parameterized policy, the policy gradient theorem is used to maximize the target function $J(\theta)$ relevant to the policy [24]

$$\begin{aligned}\nabla_\theta J\ (\theta) &= \sum_s p_{\pi_\theta}\ (s)\ \sum_a Q_{\pi_\theta}\ (s,a)\ \nabla_\theta \pi_\theta\ (a|s) \\ &= \mathbb{E}_\pi\ \left[\nabla_\theta \log \pi_\theta\ (a|s)\ Q_{\pi_\theta}\ (s,a)\right] \end{aligned} \tag{11}$$

where $p_\pi\ (s) = \sum_t \gamma^t p\ (\ s_t = \ s|s_0,\ \pi)$, $Q_{\pi_\theta}(s,a)$ is the state-action value function, $\nabla_\theta J(\theta)$ is the gradient of the policy. Based on this theorem, θ is optimized to maximize the expected return in a policy gradient algorithm

$$\theta^* = \ \arg\max_\theta \mathbb{E}\left[\sum_t \gamma^t r_t\ [s_0,\ \pi_\theta]\right]. \tag{12}$$

In our framework, we use an actor-critic structure and a proximal policy optimization (PPO) method to maximize the target function [29]

$$J_{\text{PPO}}\ (\theta) = \sum_{t=1}^{T} \frac{\pi_\theta\ (a_t|s_t)}{\pi_{old}\ (a_t|s_t)} A_t - \gamma KL\left[\pi_{old}|\pi_\theta\right] \tag{13}$$

where KL is scatter to penalize the deviation of the distribution between old and new policies, and $A_t(s_t,\ a_t)$ is the advantage to evaluate the action. The parameterized policy is a multilayer perceptron (MLP) model with three 128-D fully connected layers and activation layers.

After the policy is pre-trained, the IRL method is used to introduce expert knowledge further and enhance the transferability. IRL seeks to learn the parameterized reward function $r_\varepsilon(s,a)$ given a set of demonstration $\mathcal{D} = [\tau_1, \ldots, \tau_n]$. τ is the entire trajectory

$$\tau\ = \{\mathcal{S}, \mathcal{A}\}\ =\ \{s_1, a_1, s_2, a_2, \ldots, s_t, a_t\}. \tag{14}$$

A maximum entropy IRL gives the policy by assuming the demonstration can be generated by the optimal policy instead of giving the reward function directly, and the optimization can be defined as [30]

$$\max_\varepsilon J\ (\varepsilon) = \max_\varepsilon \mathbb{E}_{\tau\sim\mathcal{D}}\left[\log p_\varepsilon\ (\tau)\right] \tag{15}$$

where p_ε is the state visitation frequency and defined as $p(s_0)\prod_{t=0}^{T} p(s_{t+1}|s_t,\ a_t)e^{r_\varepsilon(s_t,\ a_t)}$, and r_θ is the learned parameterized reward function. Further, the gradient calculation with respect to ε can be defined as

$$\begin{aligned}\nabla_\varepsilon J\ (\varepsilon) &= \mathbb{E}_\mathcal{D}\ \left[\nabla_\varepsilon \log p_\varepsilon\ (\tau)\right] \\ &= \mathbb{E}_\mathcal{D}\ \left[\sum_{t\ =\ 0}^{T} \nabla_\varepsilon r_\varepsilon\ (s_t,\ a_t)\right] \\ &\quad - \mathbb{E}_{p_\varepsilon}\ \left[\sum_{t\ =\ 0}^{T} \nabla_\varepsilon r_\varepsilon\ (s_t,\ a_t)\right] \\ &= \sum_{t=0}^{T} \mathbb{E}_\mathcal{D}\left[\nabla_\varepsilon r_\varepsilon\ (s_t,\ a_t)\right] - \mathbb{E}_{p_{\varepsilon,t}}\left[\nabla_\varepsilon r_\varepsilon\ (s_t,\ a_t)\right]. \end{aligned} \tag{16}$$

From (15), the calculation of p_ε depends on the optimal policy and the uncertain environmental dynamics $p(s_{t+1}|s_t,\ a_t)$. Finn et al. [26] proposed a guided cost learning method to replace the optimization objective with an estimate between the demonstration sequence τ and the sequence generated by RL policy $\pi_\varepsilon\ (\pi_\theta)$, and defined the process as an adversarial generative

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network (GAN) optimization process

$$\nabla_\theta J(\theta) \approx \underbrace{\sum_{t=1}^{T} \mathbb{E}_{\mathcal{D}}\left[\nabla_\varepsilon r_\varepsilon(\tau)\right]}_{\text{Demonstration}} - \underbrace{\log \pi_\varepsilon(\tau)}_{\text{RL policy}}$$
$$\approx \underbrace{\sum_{t=1}^{T} \mathbb{E}_{\mathcal{D}}\left[\nabla_\varepsilon r_\varepsilon(\tau)\right]}_{\text{Demonstration}} - \underbrace{\sum_{t=1}^{T} \mathbb{E}_{\pi}\left[\frac{p_\varepsilon(\tau)}{\mu_\varepsilon(\tau)} \nabla_\varepsilon r_\varepsilon(\tau)\right]}_{\text{RL policy}} \tag{17}$$

where μ_ε is separate importance sampling distribution trained from the demonstrations. The policy network in the generator is updated according to the learned rewards, resulting in alternate optimization of the parameterized reward function and the policy network. In a GAN-IRL process, a discriminator D can be formulated as

$$D_\theta(\tau) = \frac{\mathbb{E}\left(r_\varepsilon(\tau)\right)}{\mathbb{E}\left(r_\varepsilon(\tau)\right) + \pi_\varepsilon(\tau)}. \tag{18}$$

This is an adversarial process between the generated actions and the demonstration. The policy $\pi_\theta(\tau)$ can be optimized by:

$$r(\tau) = \log\left(1 - D_\varepsilon(\tau)\right) - \log D_\varepsilon(\tau). \tag{19}$$

In our proposed single-step control strategy for RUS, the complete trajectory τ cannot be generated before scanning. More importantly, this article seeks to achieve probe control in an unstructured environment, where the entire trajectory is complicated to record, especially in an actual clinical application, where the surface stiffness and curvature vary largely between different patients. Therefore, we try to obtain a force-dependent learned reward to replace the simple predefined reward $\mathcal{R}$. Fu et al. [27] proposed AIRL by directly replacing the complete trajectory τ with a state-action pair. The basic logic of AIRL is to provide a state-dependent reward function by adversarial training of the discriminator with expert demonstrations

$$D_\varepsilon(\tau) = \frac{\mathbb{E}\left(r_\varepsilon(s_t, a_t)\right)}{\mathbb{E}\left(r_\varepsilon(s_t, a_t)\right) + \pi_\varepsilon(s_t, a_t)}. \tag{20}$$

The gradient of the objective function for the discriminator and the objection function for policy optimization can be rewritten as

$$\nabla_\varepsilon J(\theta) = \underbrace{\sum_{t=1}^{T} \mathbb{E}_{\mathcal{D}}\left[\nabla_\varepsilon r_\varepsilon(s_t, a_t)\right]}_{\text{Demonstration}} - \underbrace{\sum_{t=1}^{T} \mathbb{E}_{\pi}\left[\frac{p_\varepsilon(s_t, a_t)}{\mu_t(s_t, a_t)} \nabla_\varepsilon r_\varepsilon(s_t, a_t)\right]}_{\text{RL policy}} \tag{21}$$

$$r(s_t, a_t) = \log\left(1 - D_\varepsilon(s_t, a_t)\right) - \log D_\varepsilon(s_t, a_t). \tag{22}$$

Combined with our control strategy, the state consists of five continuous force observations, which means the next state is also input in the agent. We finally used the assignment normalized AIRL for the second round of optimization, which performs better in near-optimal reward learning [30]. The complete trajectory τ is replaced by the state-action-state pairs (s_t, a_t, s_{t+1}), and the cost learning objective is [30]

$$D_\varepsilon(s_t, a_t, s_{t+1}) = \frac{\mathbb{E}\left(r_\varepsilon(s_t, a_t, s_{t+1})\right)}{\mathbb{E}\left(r_\varepsilon(s_t, a_t, s_{t+1})\right) + \pi_\varepsilon(a_t|s_t)}. \tag{23}$$

The gradient of the objective function for the discriminator and the objection function for policy optimization are as follows:

$$\nabla_\varepsilon J(\theta) = \underbrace{\sum_{t=1}^{T} \mathbb{E}_{\mathcal{D}}\left[\nabla_\varepsilon r_\varepsilon(s_t, a_t, s_{t+1})\right]}_{\text{Demonstration}} - \underbrace{\sum_{t=1}^{T} \mathbb{E}_{\pi}\left[\frac{p_\varepsilon(s_t, a_t)}{\mu_t(s_t, a_t)} \nabla_\varepsilon r_\varepsilon(s_t, a_t, s_{t+1})\right]}_{\text{RL policy}} \tag{24}$$

$$r(s_t, a_t) = \log\left(1 - D_\varepsilon(s_t, a_t)\right) - \log D_\varepsilon(s_t, a_t). \tag{25}$$

The discriminator is trained to minimize the cross-entropy loss between the expert dataset and generated samples

$$\mathcal{L}(D_\varepsilon) = -\sum_{t=0}^{T} \mathbb{E}_{s_t, a_t, s_{t+1} \sim \pi_t}\left[\log D_\varepsilon((s_t, a_t, s_{t+1}))\right] - \sum_{t=0}^{T} \mathbb{E}_{s_t, a_t, s_{t+1} \sim D}\left[\log D_\varepsilon((s_t, a_t, s_{t+1}))\right] \tag{26}$$

The parameterized reward r_ε is an MLP model with three 128-D hidden layers for IRL agent training. The generator is a parameterized policy network π_ε with the same structure as in the RL method and is optimized by PPO with the same settings. In the training process, the RL policy π_θ trained by the predefined reward r from the first round will be directly transferred for IRL optimization. The expert dataset is collected from the manually adjusted RL demonstrations. The details will be presented in the following section.

C. Demonstration Collection and Training Details

The training process is performed on a natural system instead of the usual simulation environment. As shown in Fig. 4(a), a standard cubic soft phantom is horizontally fixed in the workspace during the training process. At the beginning of each episode, the ultrasound probe contacts the standard phantom in a random posture. We first use the RL model and a pre-defined reward to establish the relationship between posture and force state. After the first-round training, the probe will be controlled by the robot to be perpendicular to the phantom when it touches the surface.

Furthermore, the first-round policy is used to generate demonstrations to increase the diversity of the dataset. The specific generation process is shown in Fig. 4(b). First, the probe adjusts its posture to reach a perpendicular relationship with the surface.

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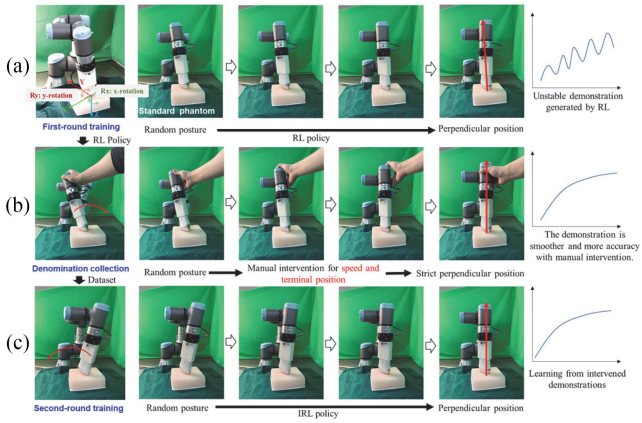


Fig. 4. Training and data generation process on the real system. A large number of demonstrations are generated from the RL agent. After the manual intervention, the velocities and probe positions in the demonstrations have more accurate and stable results. The tuned demonstrations are used to train the IRL agent.

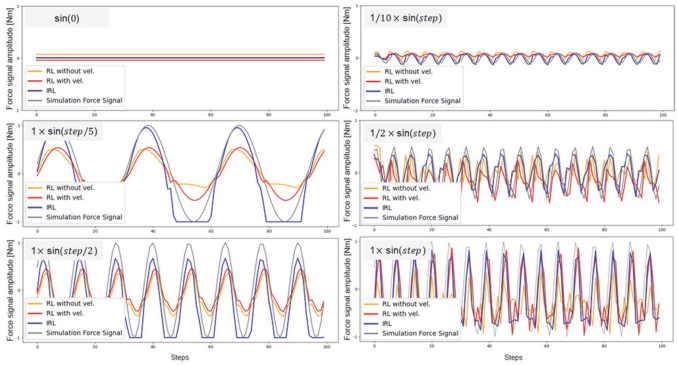


Fig. 5. Different agents were fed simulated force signals with different amplitudes and frequencies to verify the control strategy and agent response to environmental variations.

TABLE I
COMPARISON OF DIFFERENT METHODS WITH IN SIMULATION CONTACT FORCE SIGINAL

Method	Mean absolute error compared with input simulated signal (Nm) ↓				
	$\sin(t)$	$\sin(\frac{t}{2})$	$\sin(\frac{t}{5})$	$\frac{1}{2}\times\sin(t)$	$\frac{1}{10}\times\sin(t)$
RL-w/o-Vel	0.638±0.424	0.406±0.249	0.381±0.211	0.292±0.156	0.074±0.059
RL-w-Vel	0.418±0.264	0.393±0.230	0.325±0.117	0.270±0.161	0.062±0.047
IRL+DEMO	0.361±0.264	0.362±0.181	0.199±0.192	0.169±0.138	0.044±0.029

However, based only on the naïve predefined reward, the probe cannot strictly get the perpendicular position in the real-world environment, and the speed is unstable. We performed human intervention for each set of example collections to optimize the generated data. We evaluate and adjust the generated state-action pair (s_t, a_t) according to the true position of the standard phantom. Specifically, when the probe reaches the strict perpendicular position, the episode is over. The episode is considered invalid if the probe takes too many steps before reaching the perpendicular position. Moreover, we intervene manually in the RL policy's control process and record theactual velocity as the velocity in the action space. Manual intervention of the probe provides the standard for velocity control, which helps maintain its smoothness, i.e., the stability of the acceleration and its reasonableness to generate smaller velocity near the perpendicular position and faster velocity in the start position.

Three different cubic phantoms were used to generate the examples in the demonstration collection process. Each phantom was fixed in five different postures. Five hundred episodes were recorded as expert demonstrations for the second-round training. As shown in Fig. 4(c), the training and environment settings for the second-round training process are consistent with the first-round process.

III. EXPERIMENTS AND RESULTS

A. Experiment and System Setup

The system consisted of a lightweight robot (UR3, Universal robot, Denmark), a mounted wireless linear array ultrasound probe for imaging, and a 6-DOF force sensor (FT300 Force/Torque Sensor, Robotiq, Canada) for force measurement. To prevent control errors caused by force sensor errors, the sensor was reset before the start of each scan. During the control process, the frequency of the action was set to 10 Hz to ensure that the action was completed before the following action is output. The rotation speed was limited to 0.1 rad/s to ensure safety. All network calculation was performed on a work-station computer (Intel Core i7-8700K CPU, NVIDIA RTX 3080 GPU).

B. Force-Velocity-Related Control Strategy Evaluation

To verify the effectiveness of the proposed control strategy, we first evaluated the response of the force-velocity action space to a different changing sensed contact force. A standard simulation [sin(steps)] force signal was input to the different agents with the proposed velocity-force-related action space (RL) and the RL agent without the force action space (RL-without-Vel) [20]. The action sequences responded to by the agents in the experiment were recorded. In addition, simulation signals of different amplitudes and frequencies were input to show the robustness of different agents to environmental variations. The results are shown in Fig. 5 and Table I. The quantitative results indicated that the agent with velocity-force-related action space had fewer large errors and better stability when the amplitude and frequency of the input signal increased. The same validation was performed between the IRL agent with manually adjusted demonstration (IRL+DEMO) and the RL agent. Table I and Fig. 5 show that the IRL had a stable response and accurate output for different input force signals. In the evaluation of significant variation input signal [sin(t)] and fast variation input signals, the IRL + DEMO results (0.361 ± 0.264) were better than the RL (0.418 ± 0.264). Moreover, IRL + DEMO (0.199 ± 0.192) had a significant advantage over RL (0.325 ± 0.117) in evaluating input signals closer to the natural environment.

To further validated the performance of the proposed control method on the real RUS, evaluation experiments were performed on the standard cubic phantom. The experimental procedure was the same as the training procedure shown in Fig. 4. In

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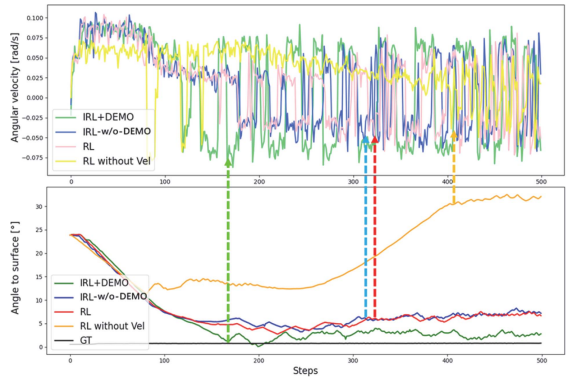


Fig. 6. Angle and angular velocity change curves in signal-step control evaluation.

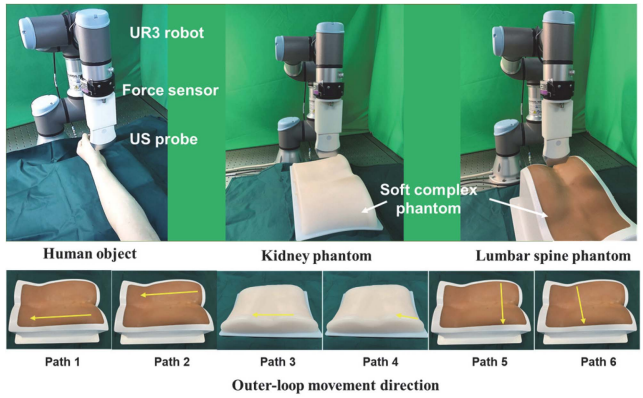


Fig. 7. Experiment and system setup. The objects being scanned were two unknown complex phantoms. Six scanning directions were predefined to perform outer-loop position control.

different test processes, the probe contacted the surface in the same posture and reached a stable posture under the control of different methods. The outer-loop position was set to zero, and the probe did not perform complete scans. In this article, all results were represented in the robot's base coordinates. As shown in Fig. 6, the angle change curve in posture indicated that the probe was adjusted at a more significant velocity at the beginning. Then, the probe reached a stable posture after a period of oscillation. Finally, the policy output a regular oscillation decision to maintain the posture. Compared with RL-without-Vel, the additional velocity action space effectively improved the control accuracy and stability. The lack of variable velocity made it challenging for the probe to reach optimal stability. Regarding posture accuracy, the IRL+DEMO was closer to the ground truth (GT) than the other methods and was significantly better than the IRL agent without manual adjustment demonstration (IRL-w/o-DEMO). The principle of IRL was to learn parameterized reward functions from demonstrations. The IRL agent without manually tuned demonstrations had no additional information introduced compared to the RL agent, which also led to similar results to RL. After reaching the stable posture, the angular error of the IRL+DEMO was 2.8°. The IRL-w/o-DEMO and the RL angular errors were 6.2° and 6.1°.

In the velocity change curve, the method with velocity action space had a variation in the early stage and a faster angular velocity at the beginning (0–100 steps). The difference was that the IRL+DEMO method reached the oscillation state earlier than the other methods. The change curves of speed and angle indicated the effectiveness of the control strategy. Compared to RL and IRL-w/o-DEMO, the introduction of manually tuned demonstrations in IRL+DEMO showed that the parameterized reward learned additional information from the demonstrations and effectively improved control stability and accuracy. *A comparison video of the single-step evaluation can be found in Movie 1 Part 1.*

C. Posture Accuracy in Continuous Scanning

As shown in Fig. 7, two soft complex phantoms (lumbar spine and kidney) with different characteristics were used as the objects in continuous scanning. All parameters and characteristics of the phantoms were unknown to the trained policy. The policies

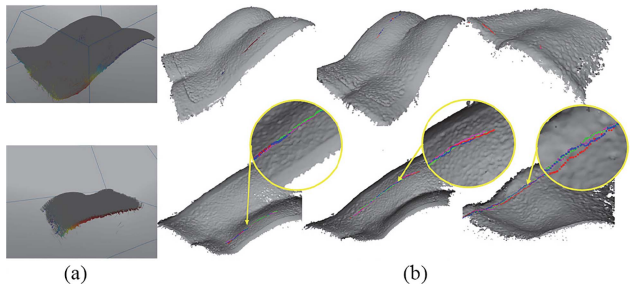


Fig. 8. (a) Visualization of the reconstructed 3-D surfaces for trajectory comparison and evaluation. (b) Overlaid static surface and scanning trajectories.

obtained by training on the standard phantom were used to perform the scanning process without any extra adjustment. Six scanning directions were prepositioned on three phantoms as the outer-loop control trajectories. The RUS moved point-to-point to collect images with different control methods for comparison. These scanning trajectories were selected with a significant morphological variation to make the task more challenging. We recorded and evaluated the contact force, the probe's postures, and the collected ultrasound images. Each scanning path was repeated ten times to calculate an average value.

We recorded the posture change results in the six trajectories to verify the continuous posture during continuous scanning. In addition, we performed manual scanning (manual) for the six trajectories for comparison. Considering the probe was not strictly perpendicular to the surface in the manual operations, a high-precision stereo camera (Ensenso-N35 stereo camera, IDS, Germany) was used to reconstruct the static surface and normal vector estimation. Due to the unpredictable surface deformation, the reconstructed 3-D model and manual operation results were used as a reference rather than the GT. As shown in Fig. 8, all paths were under the 3-D surface due to the deformation, which could not be reconstructed due to the occlusion of the probe.

Fig. 9 shows the probe's change curves in the contact direction (z-direction) in base coordinates and the Euler angles (roll and pitch) in the base coordinates during different paths. The horizontal coordinates were the direction of the movement

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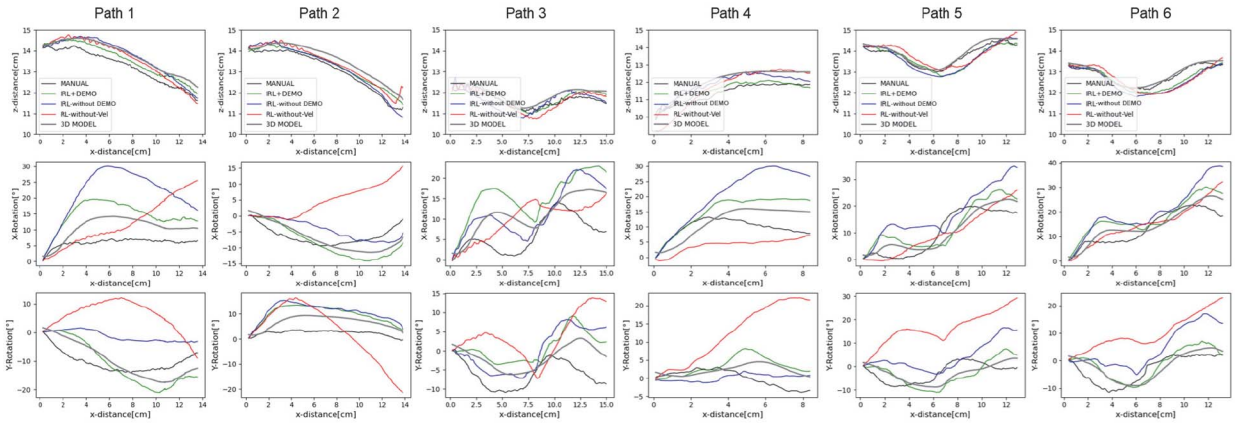


Fig. 9. Posture change curves and the contact position change curves in the six continuous scans with different methods.

TABLE II
COMPARISON OF DIFFERENT METHODS WITH MANUAL AND 3-D RECONSTRUCTED MODELS IN THE CONTACT POSITION (Z-DIRECTION)

Method	Compared with Manual [mm]/Compared with 3-D model [mm] ↓					
	Path 1	Path 2	Path 3	Path 4	Path 5	Path 6
Manual	N/A 5.4±1.3	N/A 4.9±1.8	N/A 2.6±1.5	N/A 6.6±2.4	N/A 1.2±0.9	N/A 0.8±0.6
RL -without-Vel	3.8±1.9 2.3±2.4	3.3±1.9 2.5±1.3	2.1±1.4 2.8±2.2	6.6±2.6 4.1±3.9	2.6±1.4 2.6±1.9	3.2±2.1 3.5±2.5
IRL - without - DEMO	4.7±2.4 1.4±1.6	1.8±1.1 4.2±2.4	2.5±1.6 2.9±1.7	5.1±2.1 2.0±1.2	2.5±1.5 2.8±2.1	3.0±2.1 3.5±2.3
IRL+DEMO	3.3±0.9 2.1±1.1	2.5±1.6 2.5±1.0	2.0±1.3 1.8±1.2	1.8±1.2 5.1±2.1	2.1±1.7 2.7±2.0	2.0±1.7 2.6±1.9

TABLE III
RESULTS OF FORCE CHANGE IN SIX CONTINUOUS SCANS WITH DIFFERENT METHODS

Method	Resistance force of moving direction Fy /Pressure force Fz [N]					
	Path 1	Path 2	Path 3	Path 4	Path 5	Path 6
Manual	-2.5±2.5 8.2±1.6	-1.1±1.1 6.3±1.2	-2.4±1.9 6.3±1.2	-3.8±2.8 8.5±3.7	-1.9±2.3 6.8±3.3	-0.8±2.2 5.9±4.2
RL -without-Vel	-0.1±5.3 8.3±6.7	2.6±8.4 9.8±9.6	-0.6±2.8 7.4±5.1	-3.1±12.8 4.2±1.8	-1.2±6.7 10.7±10.9	-1.1±3.5 10.7±8.7
IRL - without - DEMO	0.7±5.5 10.2±5.0	0.9±1.0 10.4±8.8	-0.1±1.1 7.0±5.0	-1.8±6.6 4.4±2.3	-2.4±5.7 9.3±6.8	-2.9±3.2 9.4±4.6
IRL+DEMO	-1.1±1.1 6.3±1.2	-0.3±0.4 8.1±3.4	-0.5±4.6 8.6±3.9	-2.5±7.4 7.1±4.4	-1.8±4.1 8.8±3.6	-1.4±2.0 7.9±1.7

distance (yaw). The z-distance change curves showed that the probe's position was under the 3-D model (grey line) because of pressing and deformation. Table II is the quantitative comparison results of the different methods with the 3-D model and the manual operation. The standard deviation (SD) values indicate the smoothness of the control process. The introduction of manual demonstration resulted in the IRL+DEMO (1.1 to 2.0 mm) having a smaller SD in the control process compared to the RL (1.3 to 3.9 mm) and the IRL-w/o-DEMO (1.2 to 2.4 mm). Compared to the static 3-D model, the z-distance difference due to pressing in the IRL+DEMO was 2.8±1.6 mm, and the difference in the manual operations was 3.6±1.4 mm. This result indicated that the deformation caused by the proposed method during the scanning process was close to the manual procedure.

The angle change curves represented the posture accuracy during the scans. Compared with manual operation, the overall average difference of the IRL+DEMO in Rx and Ry was (6.1±3.4°, 5.2±2.7°). In comparison, the result of the IRL+DEMO was closer to the 3-D model than the IRL-w/o-DEMO (3.1±1.9°, 2.4±2.3°) and the difference was (2.3±1.3°, 1.9±1.2°). The smaller difference with the 3-D model indicated that the probe could maintain an accurate angular relationship with the surface in continuous scanning. However, there was a bigger difference with manual operation. The difference between the IRL+DEMO and the 3-D model in long-distance scans (path 1 and path 2) was (2.7±1.2°, 2.4±1.3°), and in the paths with large variation (path 5 and path 6) was (2.5±1.3°, 1.5±1.0°). On the softer kidney phantom (path 3 and path 4), the difference between the IRL-DEMO and the 3-D model increased (3.8±1.4°, 2.0±1.3°). The results of these continuous scans were similar to those of previous single-step evaluations, demonstrating the feasibility of the proposed method during continuous scanning. Overall, compared to the RL and the IRL-w/o-DEMO policy, the posture generated by the policies with the learned reward was closer to the 3-D model, represented better transferability between different phantoms, and could adapt to different paths. *A comparison video of the policy can be found in Movie 1 Part 2.*

D. Force Evaluation in Continuous Scanning

Contact force was recorded as shown in Fig. 10. The x-direction was the outer-loop moving direction, and Fy represented the frictional force and the resistance generated by the deformation during the scanning process. Since the experimental phantoms were coated with enough ultrasonic coupling agents to match the real-world scenario, surface deformation was the main reason for the resistance. The contact force during the scanning process represented the controller's stability in the dynamic environment and the system's safety.

As stated in the previous posture evaluation, the deformation in the contact direction represents the relevant pressure force.

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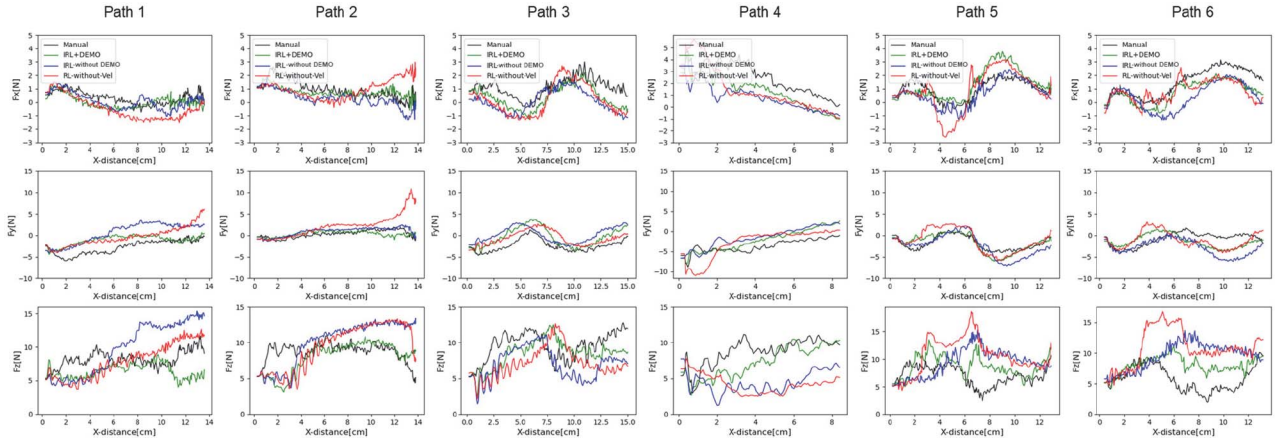


Fig. 10. Force change curves in six continuous scans with different methods.

The quantitative comparison results is shown in Table III. In all paths, the average pressure force of the IRL+DEMO was 9.0±3.6 N. The average pressure force of the RL and the IRL-w/o-DEMO were 9.8±3.7 N and 9.7±8.4 N. The smaller SD values further demonstrated the stability of the learned reward. In particular, the results of the IRL-DEMO (green line) reached the stable posture faster compared with other methods (red and blue lines) at the beginning of the scan (0 to 5 cm), which accorded with the results of the control strategy evaluation in Section III-B. The policy from the manually adjusted demonstration was more stable regarding force control. Compared to manual operation (8.0±3.1 N), the proposed method generated a more significant pressure force. In path 5 and path 6, the phantom's shape and stiffness significantly varied. The phantom's middle part (6 to 8 cm) was the bone and more complex. Thus, the automatic and manual results had opposite trends in pressure force.

Regarding Fy, the resistance force, the average value of IRL-DEMO was -1.5 ± 3.7 N. RL, IRL-w/o-DEMO, and manual averages were -1.6 ± 6.7 N, -1.6 ± 4.4 N, and -2.4 ± 2.4 N. The results showed that our method generated less resistance and deformation during the scans, further demonstrating the additional information learned in the parametrized reward and policy. Although the RL approach obtained better results in path 3, by combining the probe posture, it could be found that the probe was not in contact with the surface at the correct angle. In addition, the methods with the learned reward improved the stability of different phantoms. This also indicated that the learned reward could assist in making correct control during the scanning process, even in the absence of resistance information during the training process. *Several different autonomic scanning processes can be found in Movie 1 Part 3.*

E. Ultrasound Imaging and In-Vivo Experiment

We recorded the internal structure of the ultrasound images acquired to validate the feasibility. The images obtained in three different paths using our method are shown in Fig. 11, where five positions were chosen evenly from the paths. Images obtained by manual manipulation in the same positions were inserted for comparison. The ultrasound images obtained by our method had

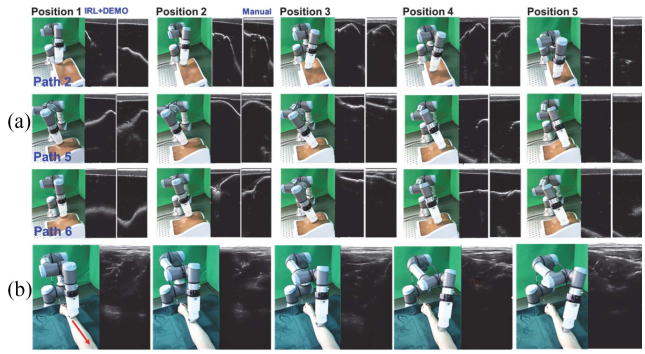


Fig. 11. (a) Comparison of ultrasound images. (b) Scanning process for a random human target.

a similar structure as those manually obtained in the middle of the process. The reason for the difference at position 1 was that the probe required steps to adjust the posture at the beginning. The average structural similarity of collected ultrasound images between our method and manual operation was 0.654.

We further performed human target scans using the same policy for training and testing on the phantom. A single volunteer maintained normal breathing during the experiment. The policy and controller being used did not have any extra adjustments. In the scanning processes, the probe was in contact with the skin and moved along the human surface. The probe maintained a stable connection with the skin during the scan, and the image remained complete. The scanning process of the human was approved by the local ethics committee (No. 20210074). *Several autonomic scanning processes for the human target can be found in Movie 1 Part 4.*

IV. Discussion and Conclusion

The uncertainty and complexity characteristics of the unstructured environment pose a significant challenge for stable and accurate robotic ultrasound scanning control. In this article, we propose a learning-based control strategy to establish the nonlinear relationship between the probe's posture and contact force and perform an autonomic scanning process based on force

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information only. First, we propose a force-velocity-related action space and control strategy to deal with unpredictable deformation. Further, to introduce expert knowledge and simplify the expert dataset collection, a hybrid policy optimization was proposed. Several experimental results indicated that our method could perform stable, smooth, and continuous scanning of different and unknown soft surfaces. Our approach showed significant advantages and better transferability in different and unstructured environments.

One limitation of our approach was the rapid movement of the target. In this article, the experimental subjects were static or slightly moving (such as breathing). This scenario was generally consistent with the actual clinical setting. However, when the scanned target was moving rapidly, the rapid change in force can lead to wrong decisions. In our future work, we will focus on the outer-loop control part of the framework. One potential solution combines force and visual information, including scene and ultrasound images. Combining multiple pieces of information can provide more accurate guidance for a targeted scanning process. In addition, since robotic ultrasound scanning is a long continuous process, learning methods for such continuous control tasks will be a topic of our future research. Overall, our method provides a simple solution for scanning RUSs without visual information and has potential applications in combination with other outer-loop trajectory generation methods.

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